Here, the unknown is the students (x). It is to be compared with the days and the length of the wall.



Step 1:

Here, less days means more students. So, it is in inverse proportion.

$\therefore$ The multiplying factor is $\frac{56}{27}$.

Step 2:

Also, less length means less students. So, it is in direct proportion.

$\therefore$ The multiplying factor is $\frac{160}{448}$.

Step 3:

$$\therefore x = 81 \times \frac{56}{27} \times \frac{160}{448}$$

$x = 60$ students.

Formula Method

Here, $P_1 = 81$, $D_1 = 56$ and $W_1 = 448$ and $P_2 = x$, $D_2 = 27$ and $W_2 = 160$

Using the formula, $\frac{P_1 \times D_1}{W_1} = \frac{P_2 \times D_2}{W_2}$

$$\Rightarrow \frac{81 \times 56}{448} = \frac{x \times 27}{160}$$

$$\Rightarrow x = \frac{81 \times 56}{448} \times \frac{160}{27} \Rightarrow x = 60 \text{ students.}$$

Example 4.22 **(Inverse – Inverse Variation)**

If 48 men working 7 hours a day can do a work in 24 days, then in how many days will 28 men working 8 hours a day can complete the same work?

Solution:

Multiplicative Factor Method

Men	Hours	Days
48	7	24
I	I	I I
28	8	x

Here, the unknown is the days (x). It is to be compared with the men and the hours.

Step 1:

Here, less men means more days. So, it is in inverse proportion.

$\therefore$ The multiplying factor is $\frac{48}{28}$.

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